

Lab 3 - Mandelbrot Loop Scheduling

Image: 640 x 360 pixels, maximum 500 iterations. Policies: static, dynamic, guided. P=2,4,8,16; chunk size=1,16,64,256; three trials per configuration. The manual lists 1920 x 1080 and 1000 iterations; this run used the smaller settings shown here.

Mean elapsed time in milliseconds (rows=P, columns=chunk size):

Static

P / C	1	16	64	256
2	17.27	15.80	17.56	29.24
4	8.29	8.40	17.47	28.93
8	5.98	7.34	17.50	29.27
16	5.98	7.56	17.60	28.86

Dynamic

P / C	1	16	64	256
2	15.33	15.94	17.74	29.49
4	9.25	8.61	17.64	29.38
8	5.78	7.25	17.70	29.43
16	5.46	7.53	17.64	29.55

Guided

P / C	1	16	64	256
2	16.07	15.85	17.66	29.38
4	8.64	8.42	17.71	29.41
8	5.53	7.20	17.69	29.39
16	5.72	7.18	17.63	29.44

Load imbalance and checksum values for every run are in lab3-results.csv; configuration means are in lab3-summary.csv.